

**Hello,
I'm Shubham Pathneja**

Ex-Robotics Intern @IIT MANDI | Team Lead @SAARTHI |
Robotics and AI TIET'27 | ROS | Computer Vision |
Robotics | Simulations | Control Systems | AI/ML/DL |



**EXPERIENCE AND
PROJECTS MEDIA**

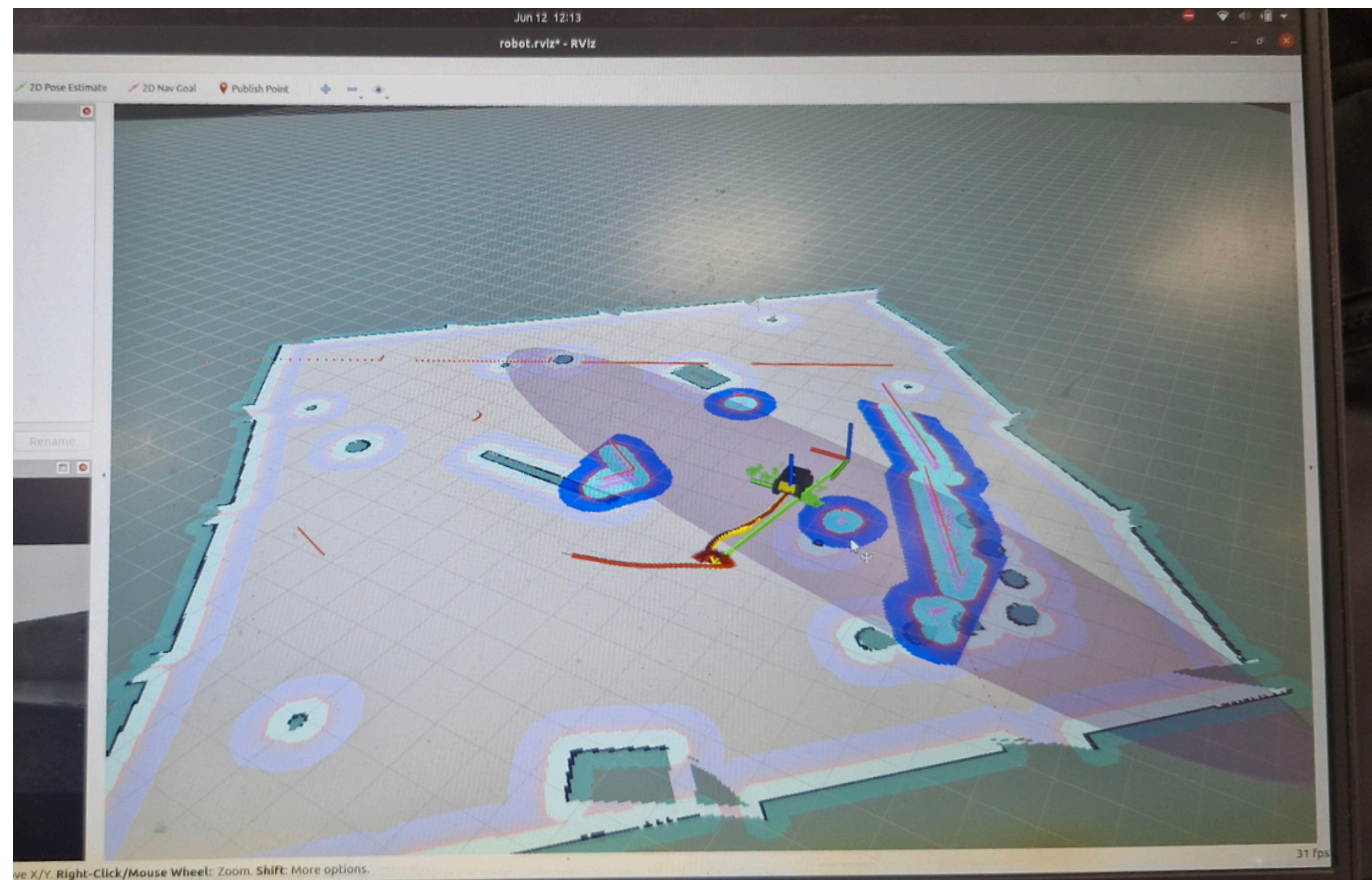


Robotics Intern @IIT Mandi

May 2026 - July 2026

LAST MILE DELIVERY Problem statement

Designed and implemented an end-to-end autonomous navigation and perception pipeline for the **Clearpath Husky A200** mobile robot.



Deployed on a **Jetson AGX Orin** to enable real-time performance using CUDA's parallel processing.



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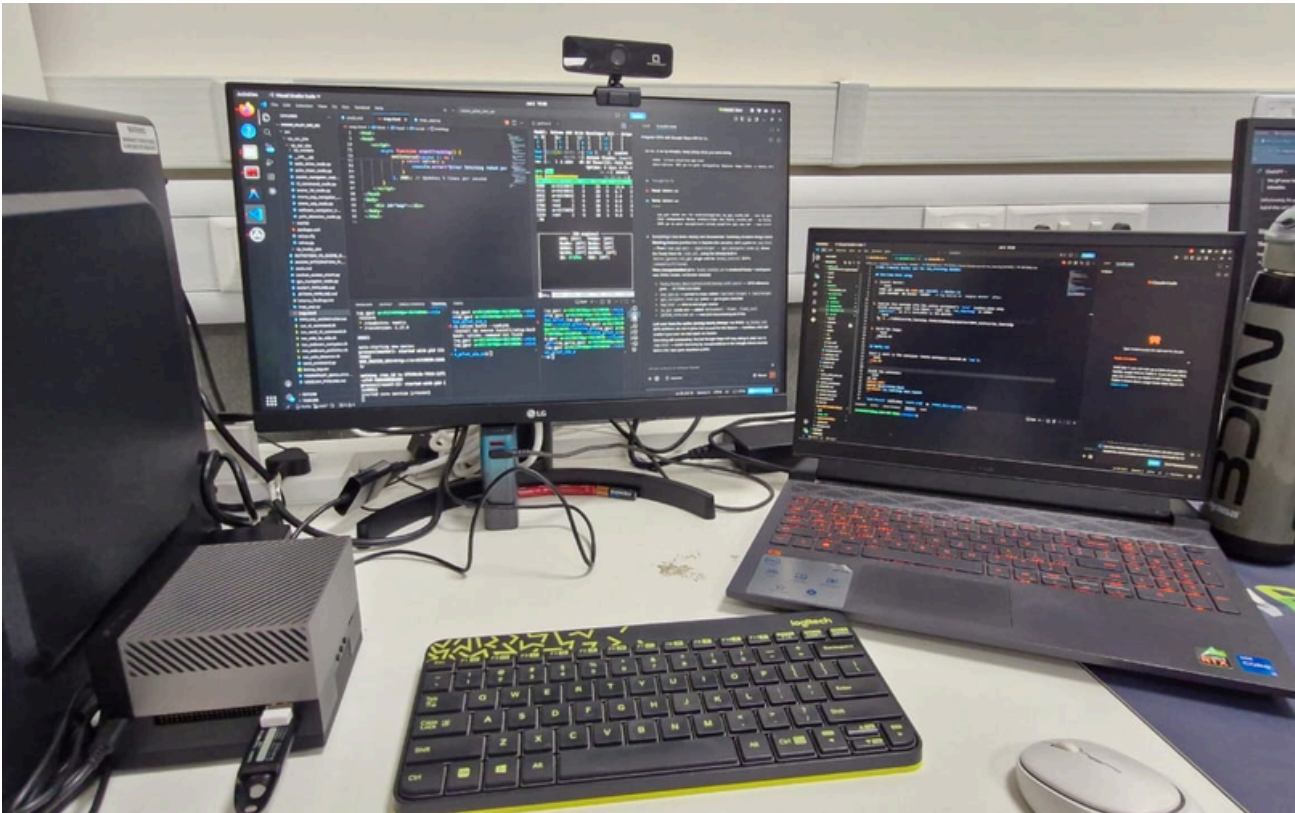
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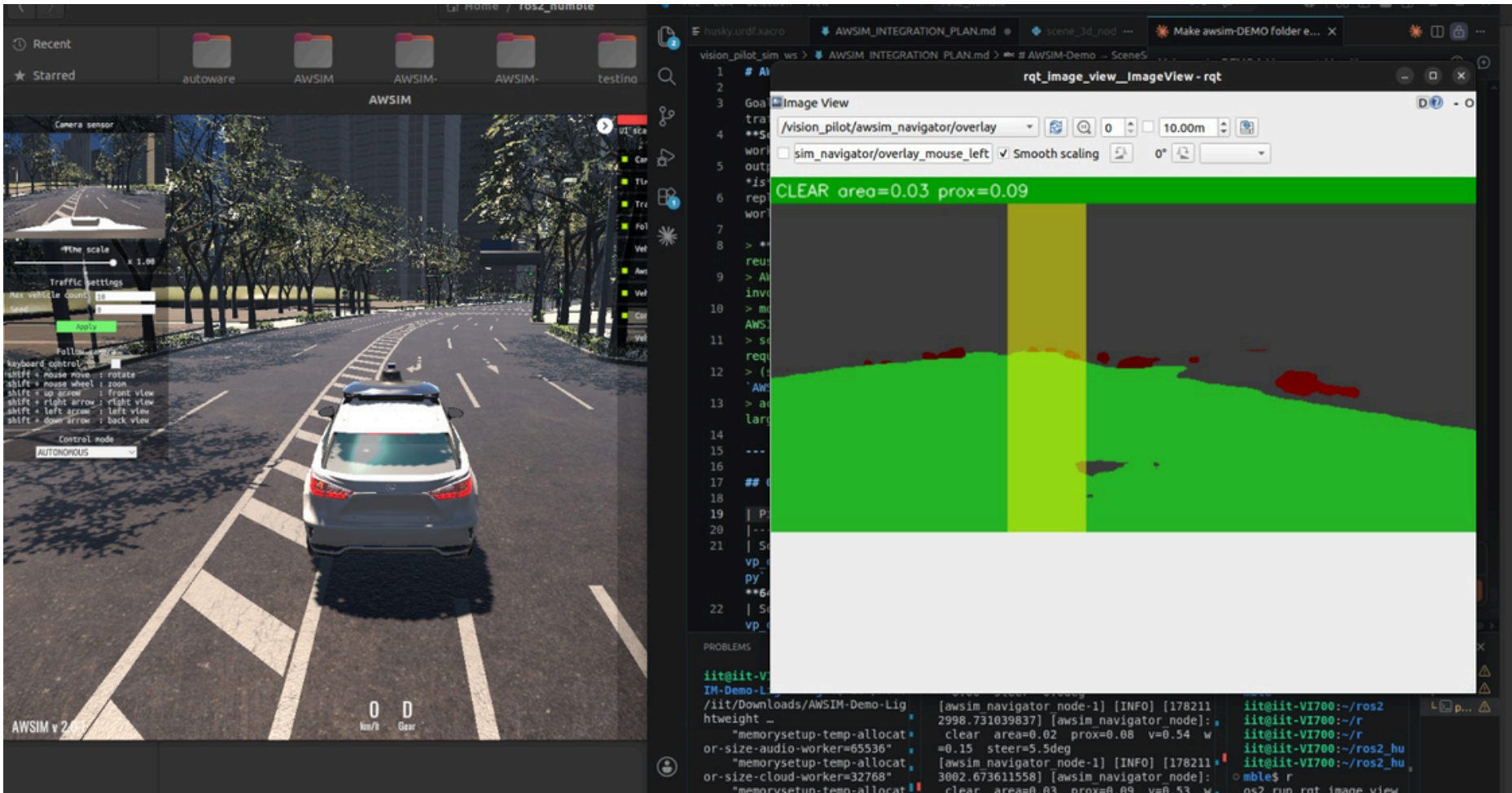
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Integrated Autoware's perception and planning stack with NVIDIA's TensorRT-optimized inference



Tested the base driving simulations using the **AWSIM** Simulator.



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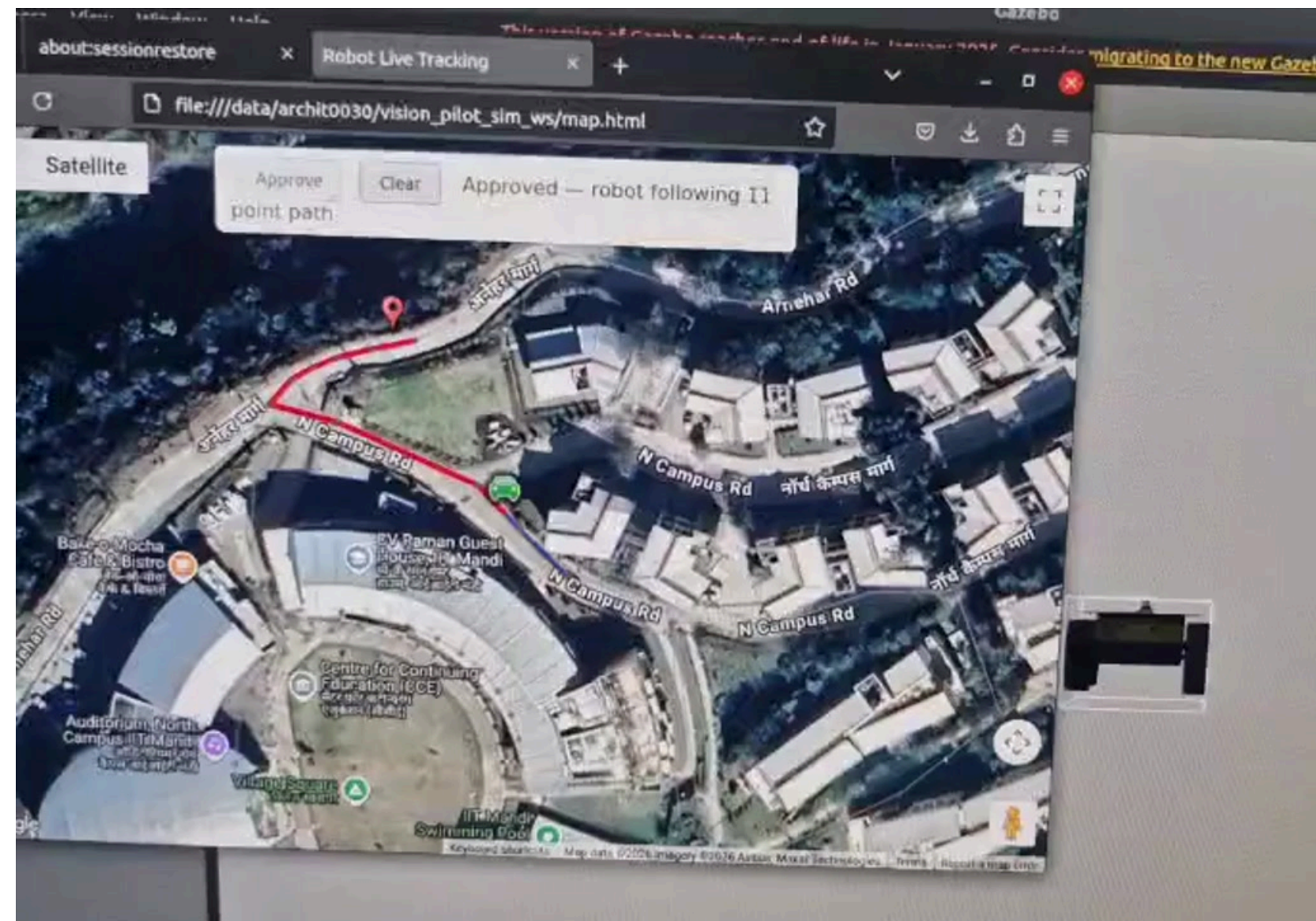
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Built a GPS-based localization and tracking module that interfaced with a host website to stream and visualize the robot's live position, densifying the path to translate GPS way-points into smooth, continuous trajectories improving path navigation



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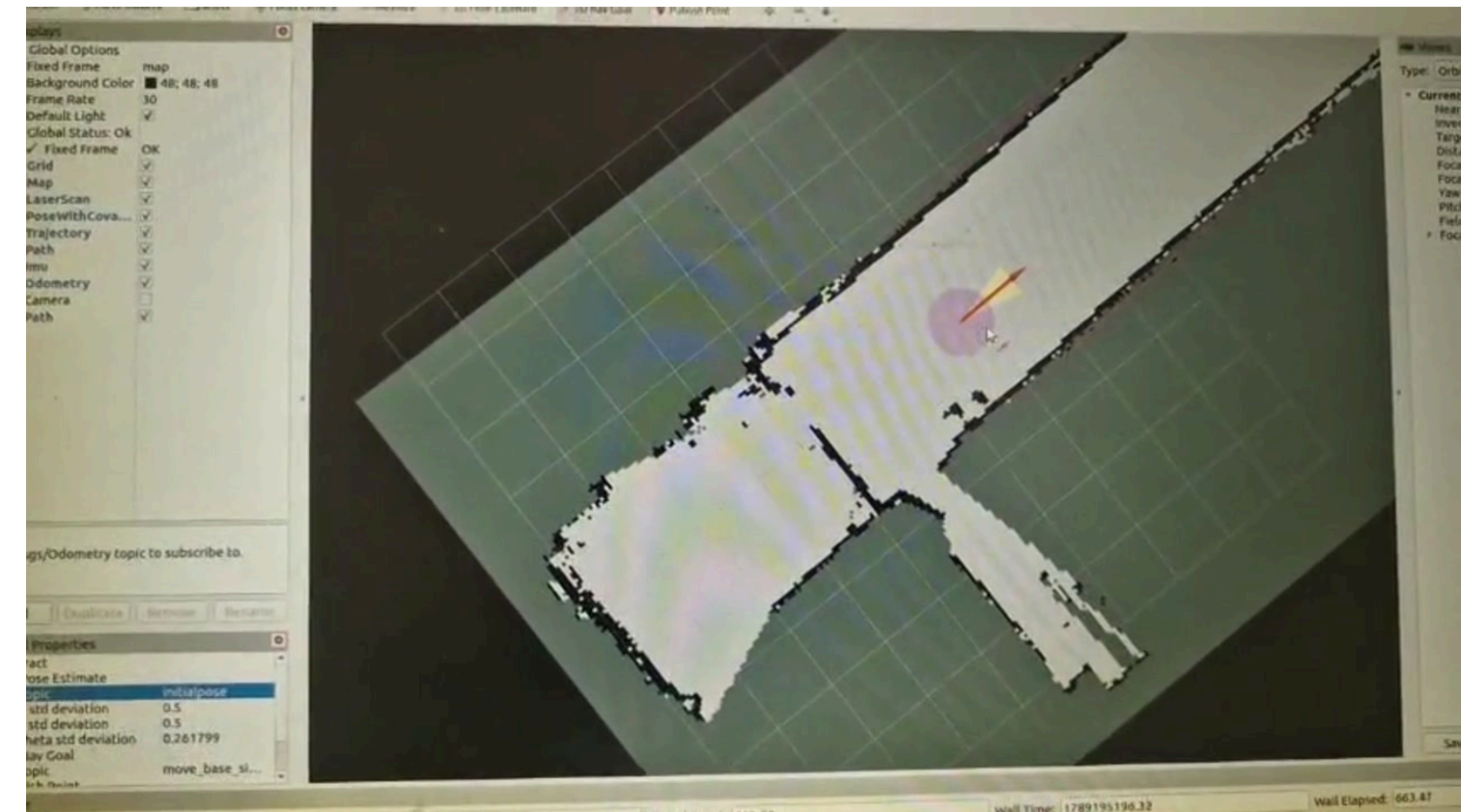
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Autonomous Exploration Bot, Robotics Lab, TIET

July 2026 - Present

Architecting an autonomous exploration and SLAM pipeline using RRT* based algorithms enabling the Differential drive Trossen Robotics **LoCoBot WX200 6DOF** to navigate through unknown indoor workspaces.



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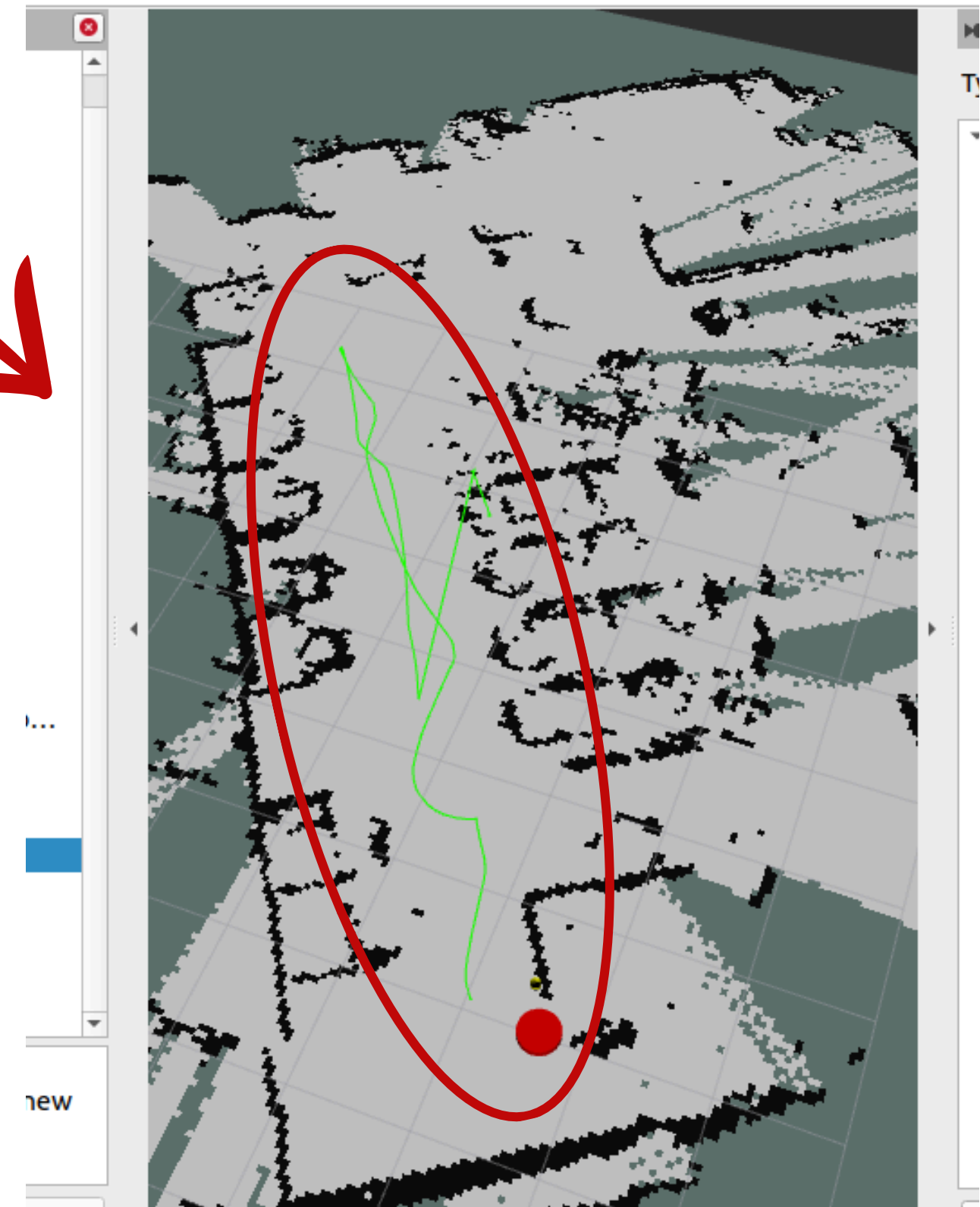
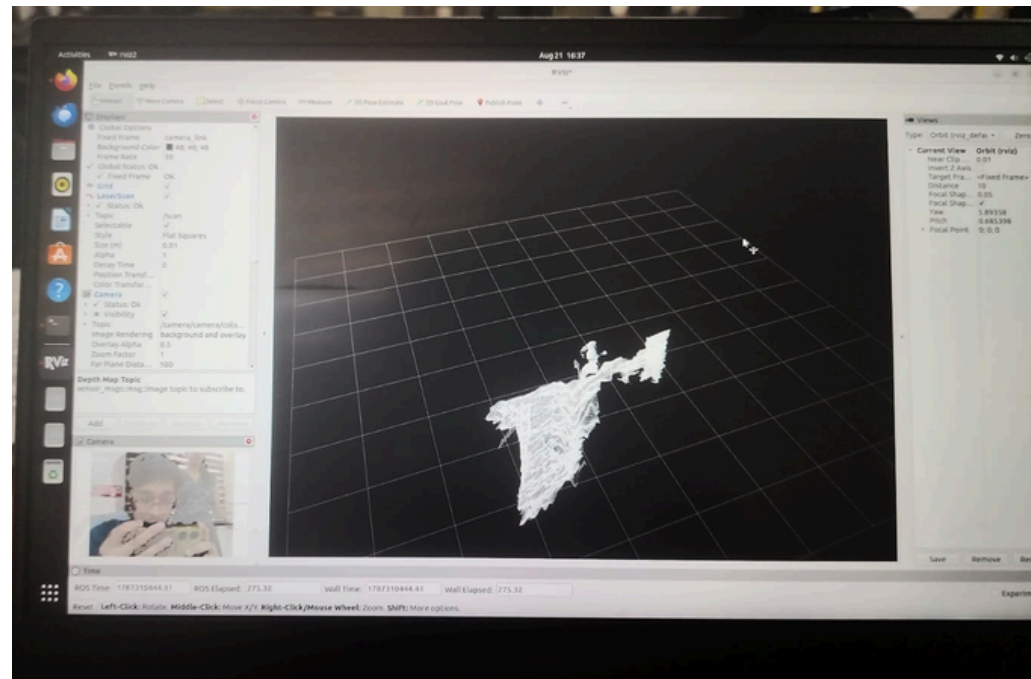
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Built real-time occupancy mapping using 2D LiDAR for large-scale spatial structure, fused with RGB-D depth camera data

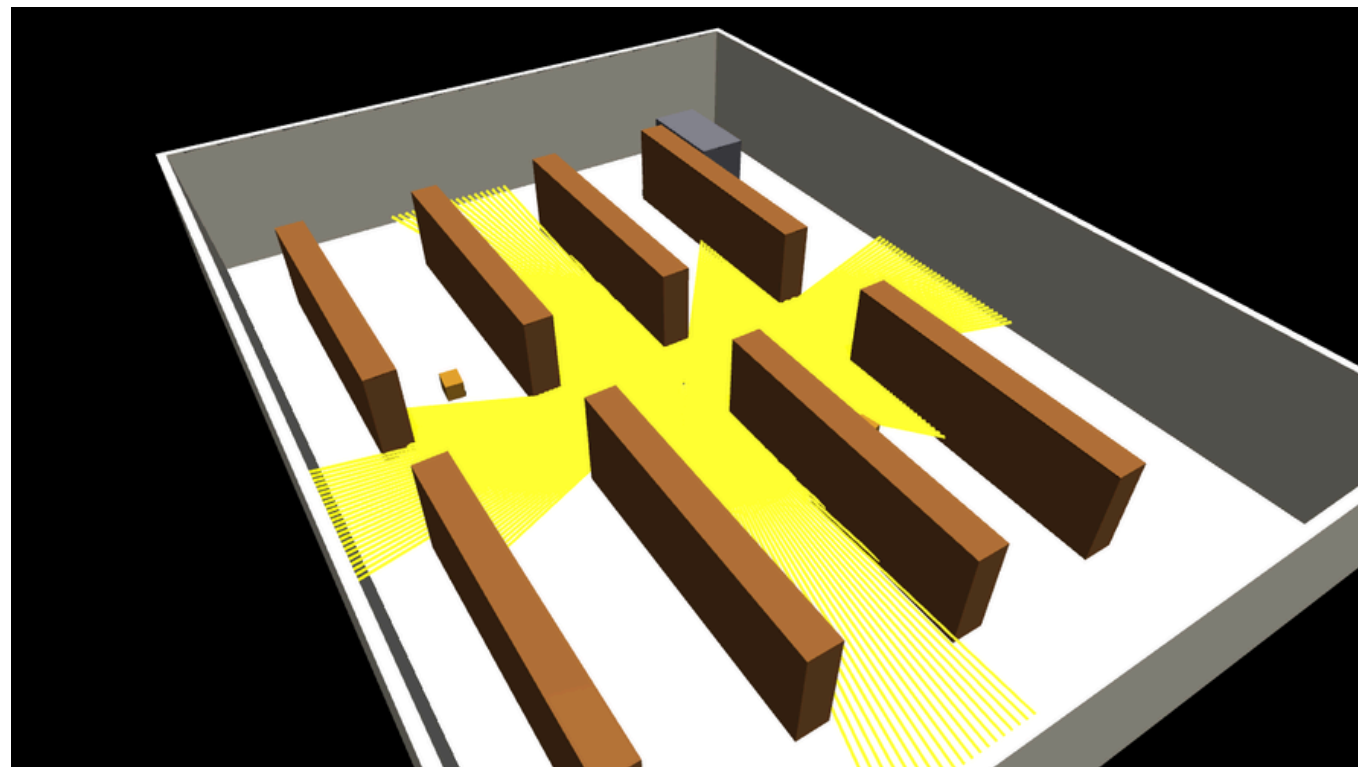


The **next step** will be towards applying the TARE exploring method from the research paper by Chao Cao, Hongbiao Zhu, Howie Choset, and Ji Zhang

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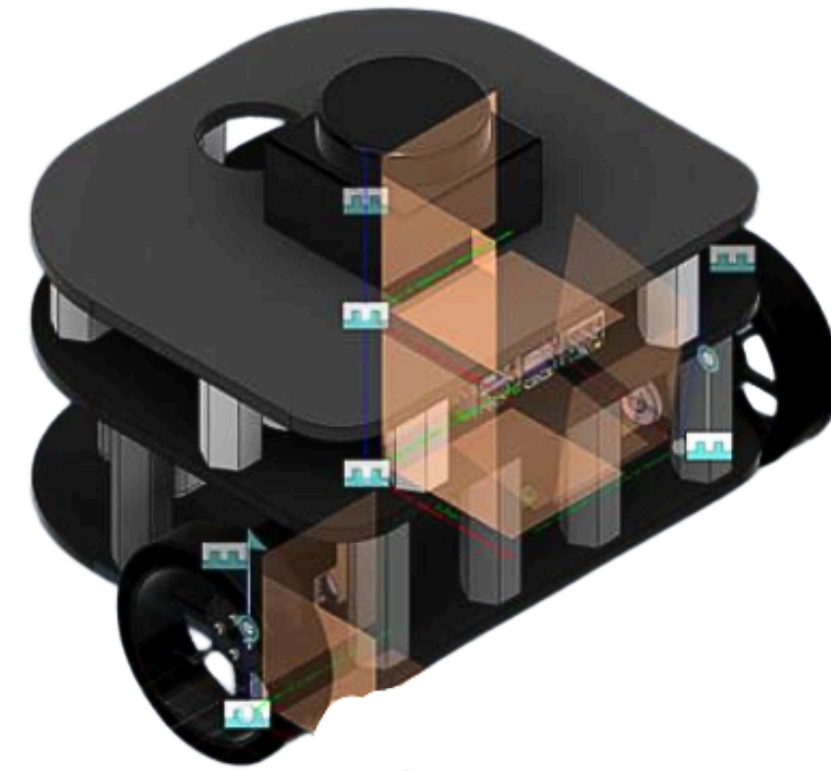
Marketplace Navigation Robot Simulator

March 2026 - June 2026



Designed and simulated a differential-drive autonomous mobile robot (AMR) in **MuJoCo**

Implemented 2D LiDAR-based SLAM to build and continuously update an occupancy map of the supermarket



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Integrated the ROS2 Nav2 stack for global path planning to target product locations and local planning



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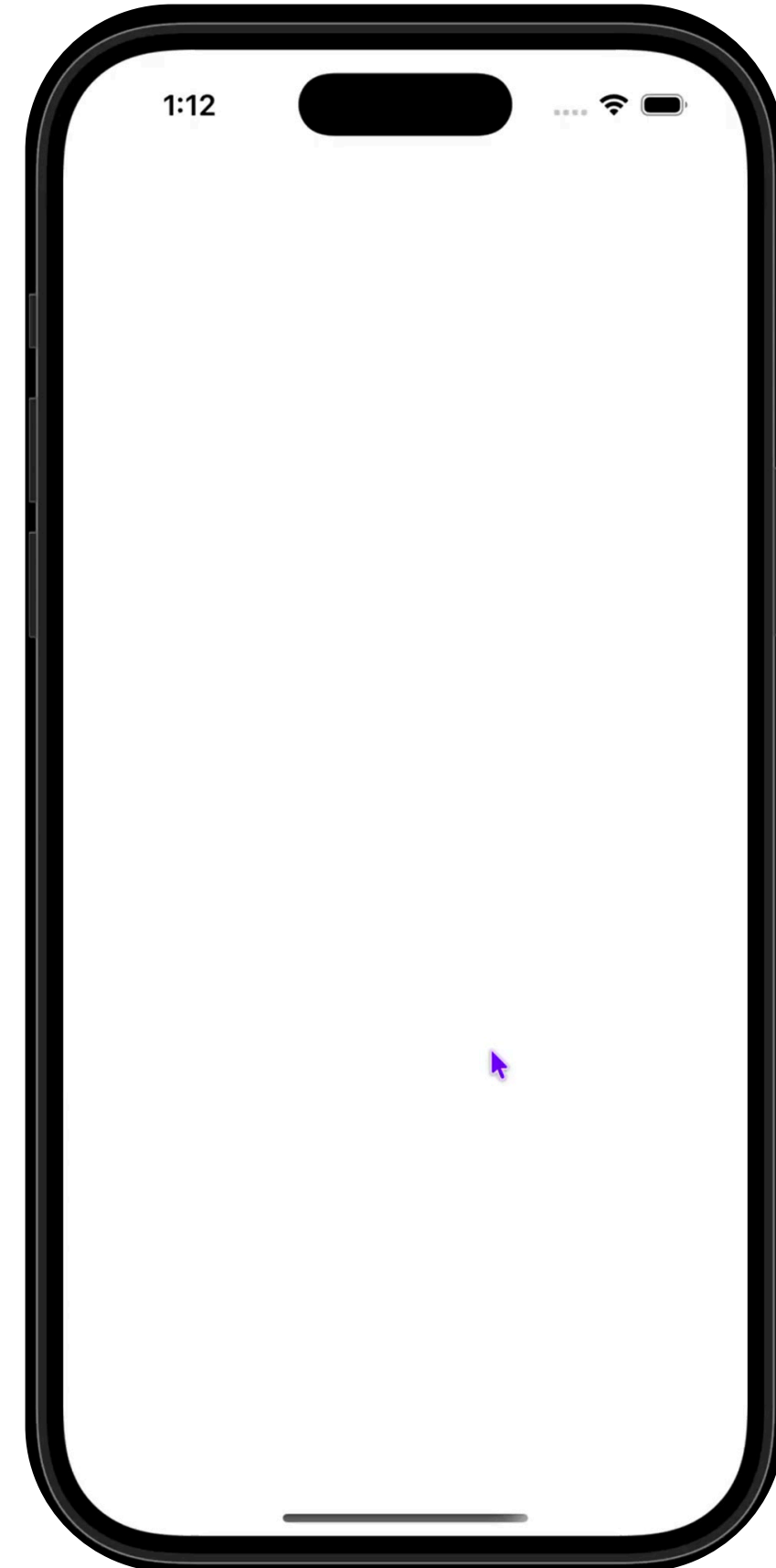
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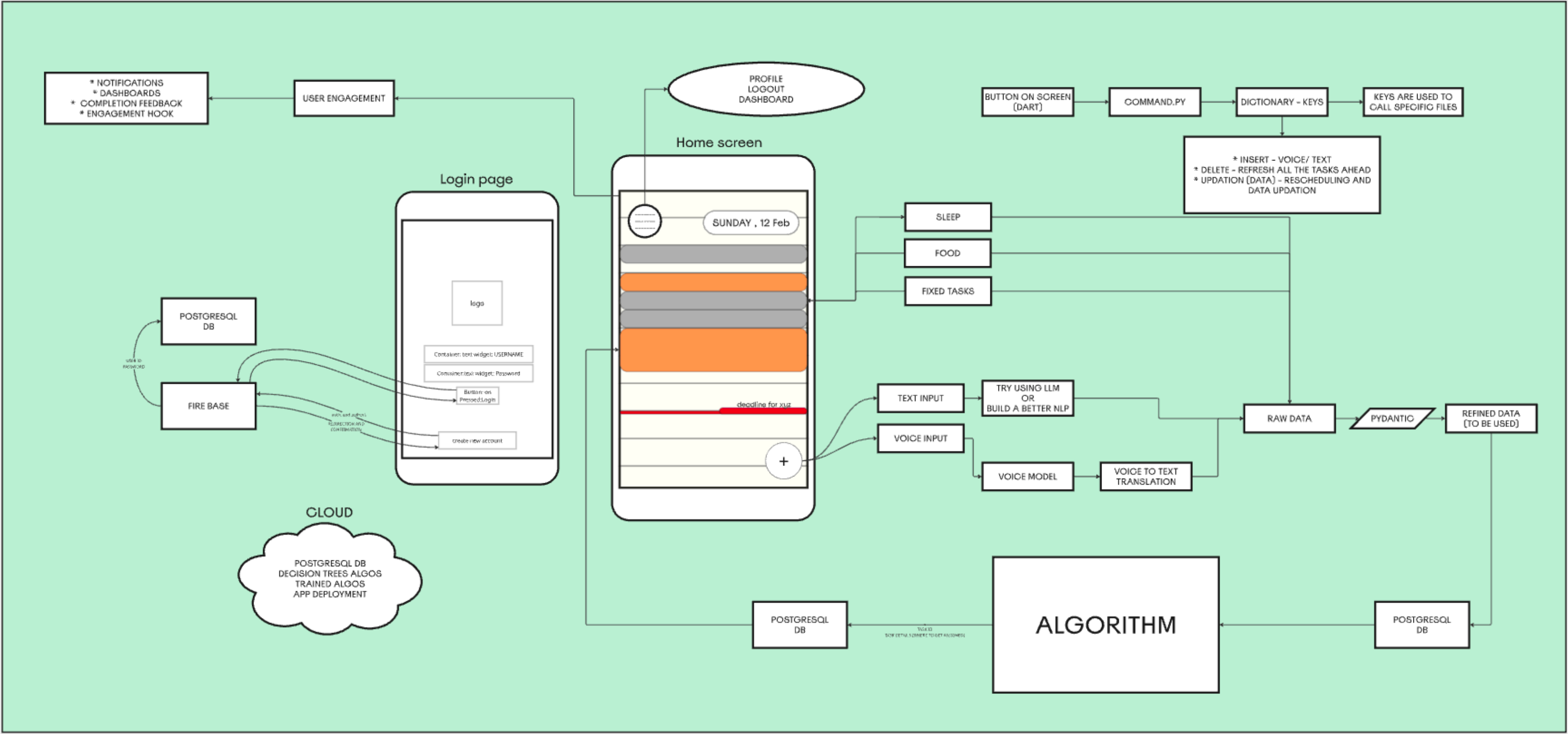


Engineering an intelligent **AI task-scheduling assistant** tailored for college students juggling coursework, deadlines, and personal commitments

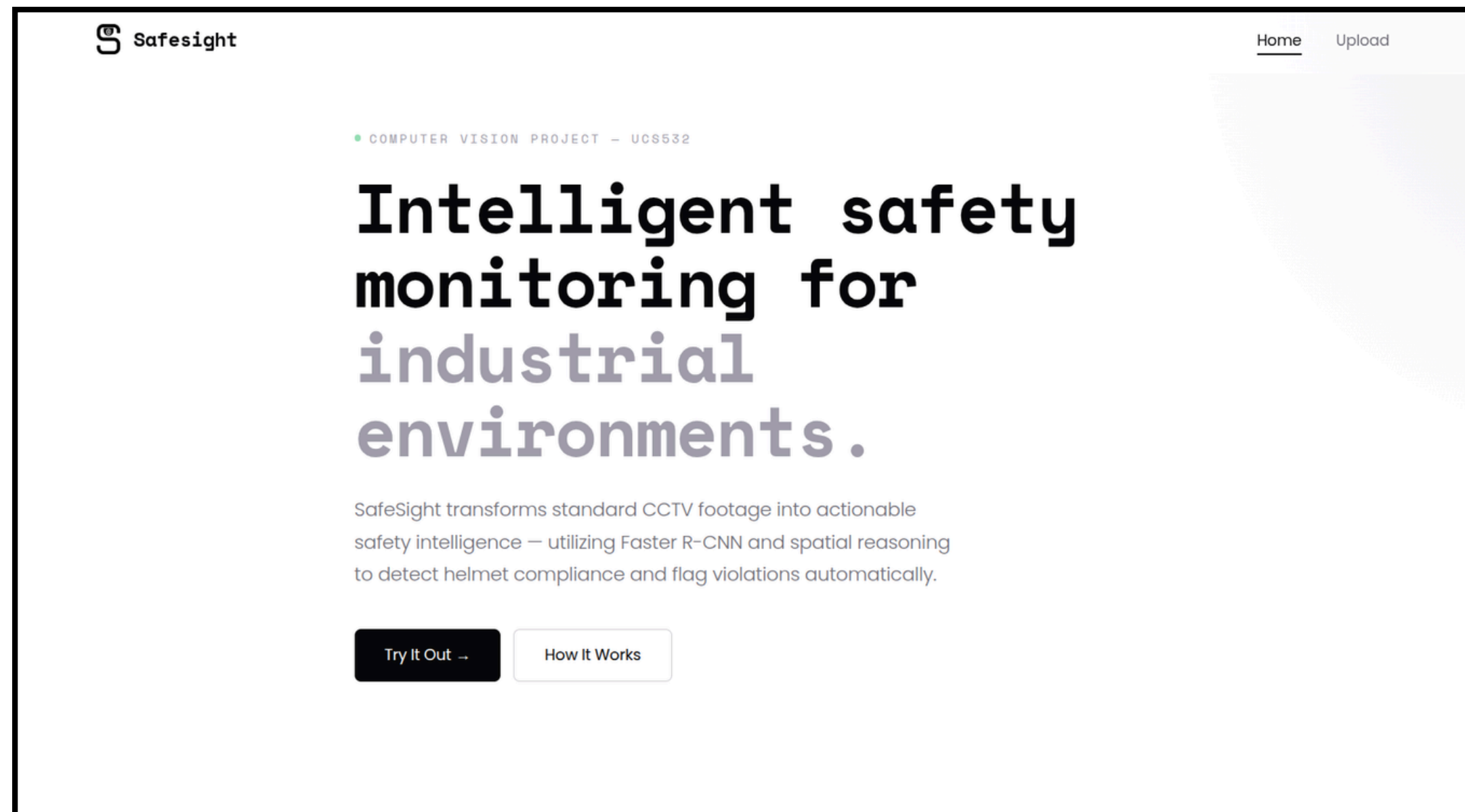
Designed a custom weighted scoring model powered by Reinforcement Learning to **learn and adapt to individual student behavior over time**



SYSTEM ARCHITECTURE DESIGN



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Built a real-time behavioral anomaly detection system deployed on existing CCTV infrastructure, using **Faster-RCNN with a ResNet50** backbone to detect and classify anomalous behavior

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It detects anomolies and also gives the violation report.

Violation Report

test_video1.mp4 Processing time: 66.7s 6 violations found

6 High

0 Medium

0 Low

TIMESTAMP	VIOLATION	SEVERITY	CONFIDENCE
00:11 - 00:12	No Helmet	High	52.5%
00:25	No Helmet	High	54.3%
00:28	No Helmet	High	66.8%
00:33 - 00:45	No Helmet	High	87.7%
00:47	No Helmet	High	80.5%
00:50 - 00:57	No Helmet	High	77.1%

Close

Export Report

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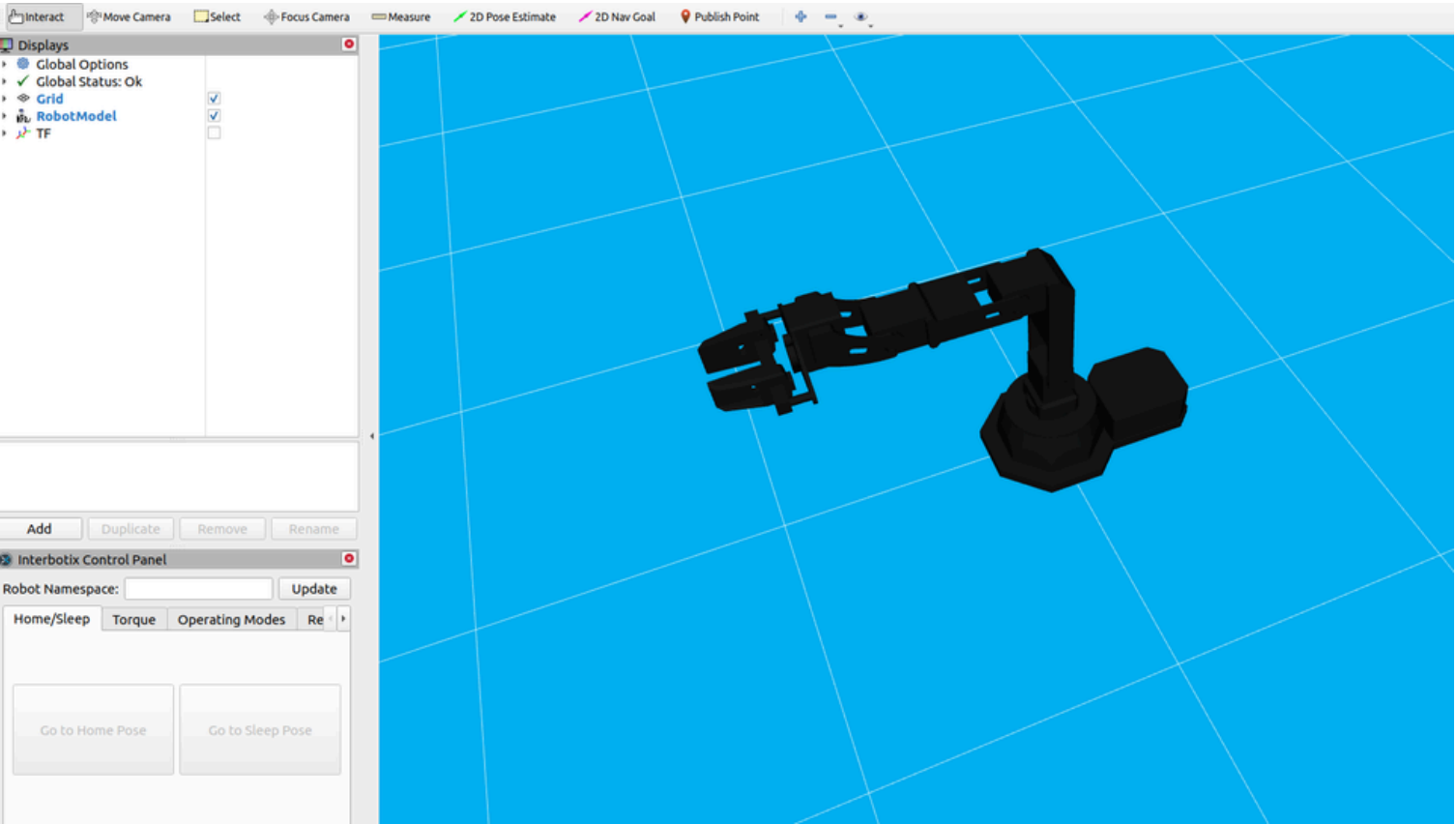
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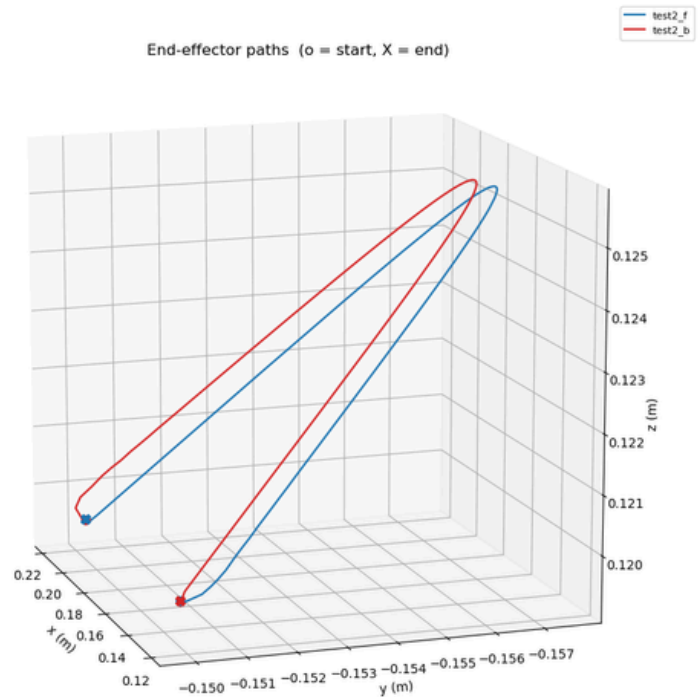
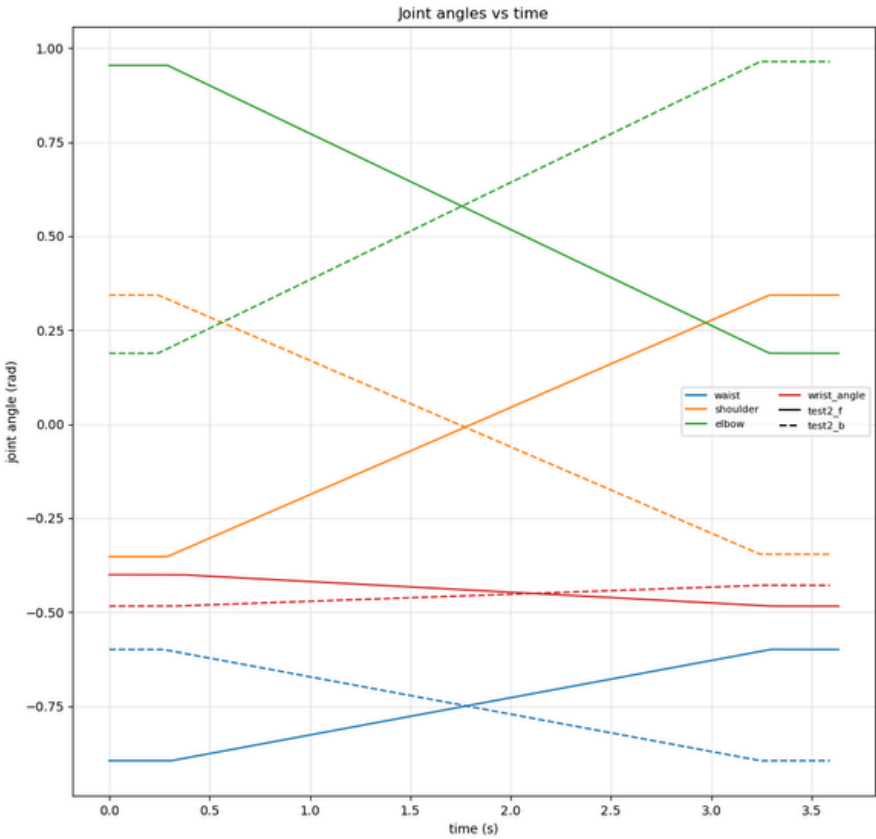
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SOME OTHER PROJECTS



Trosson Robotics LocoBot
Robotic arm for trajectory
planning and optimization.



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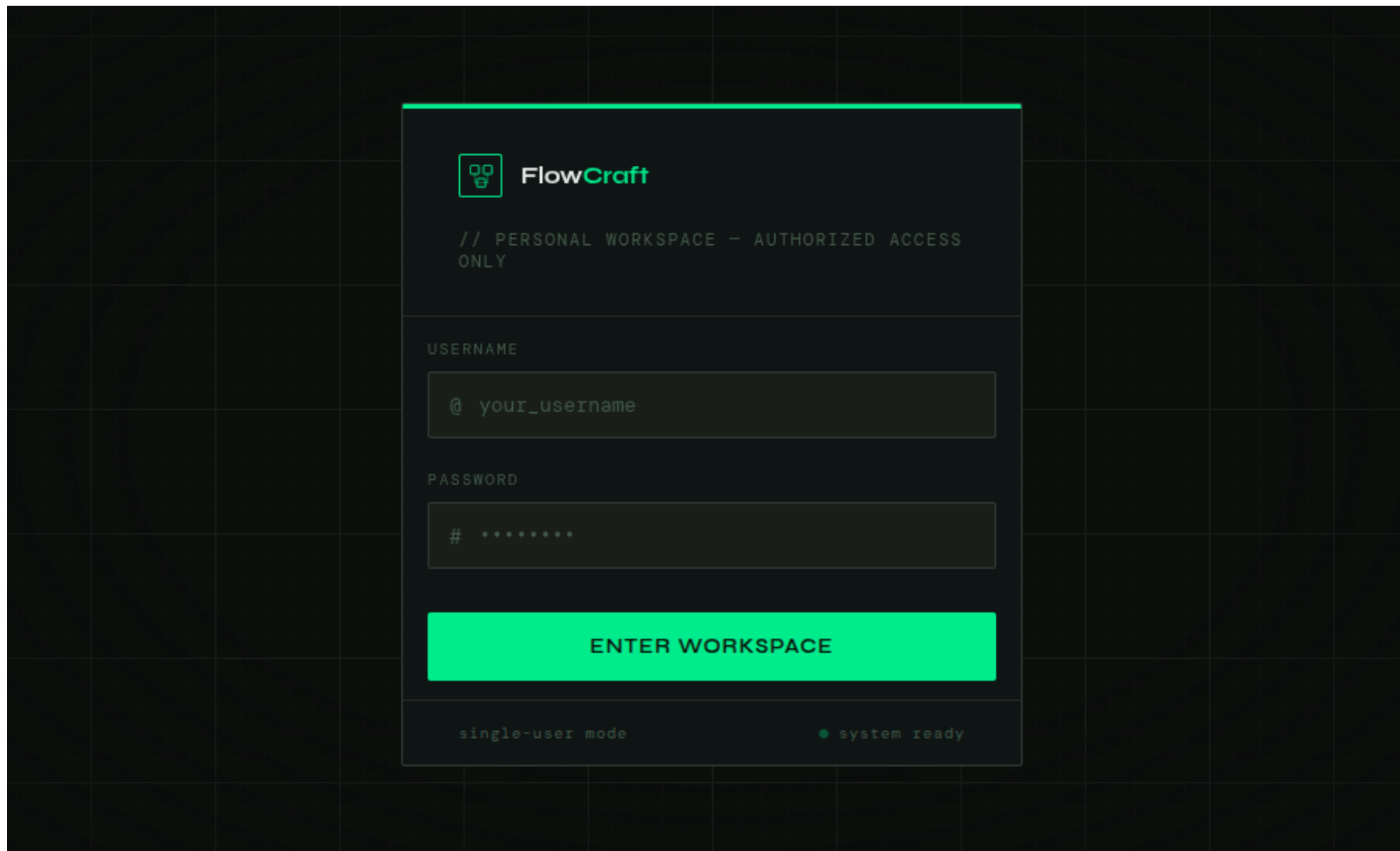
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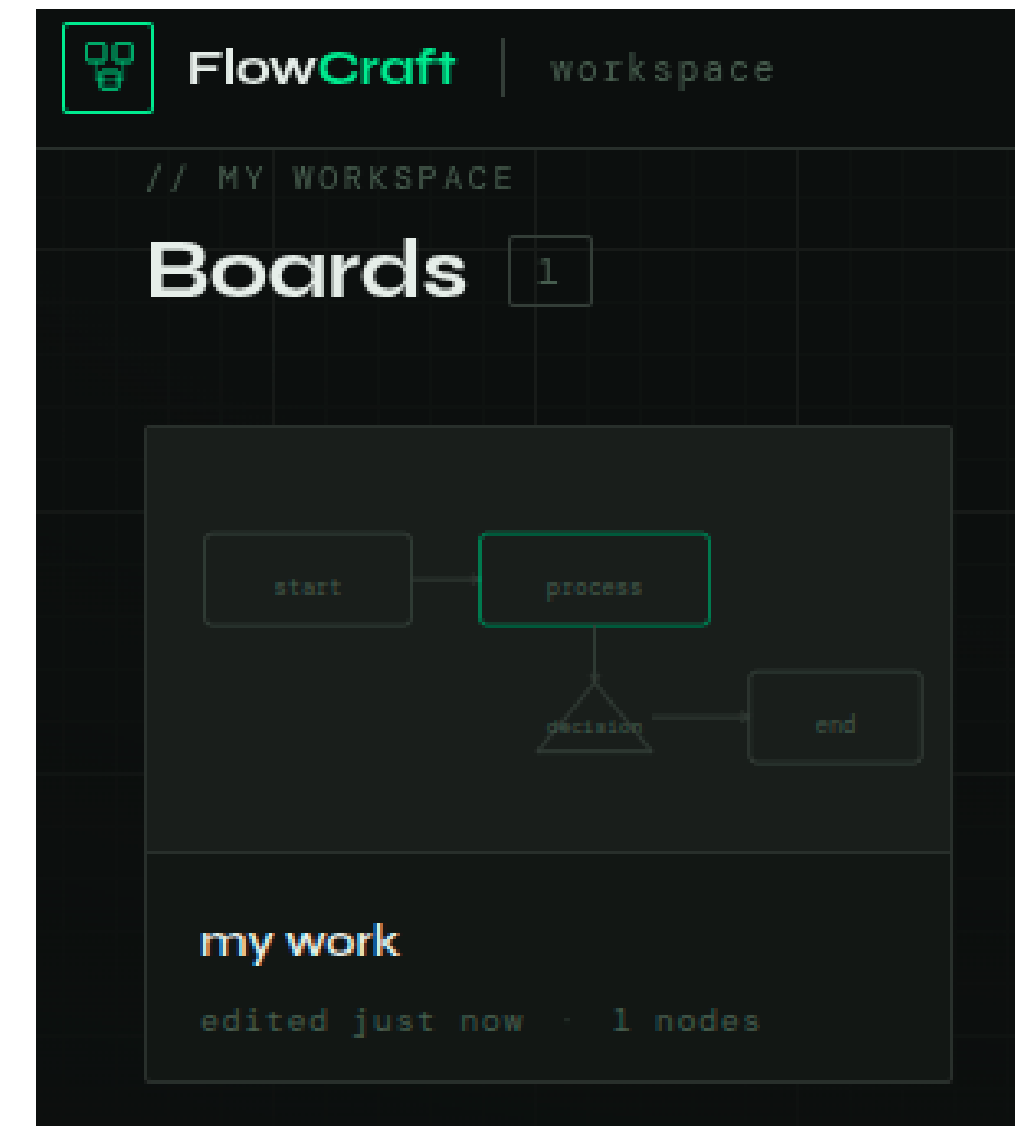
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Flowcraft (MIRO lookalike)

A **private board workspace** for the documentation of my work.



You can create your own **unlimited** boards.



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THANK YOU !!